



WeatherGPT: Viva Cheat-Sheet

[Viva Quick Read](#)

Rapid 5-Minute Revision for External Evaluation

30-Second Elevator Pitch (Memorize This!)

"WeatherGPT is an AI-powered meteorological & disaster intelligence platform that converts raw weather numbers into actionable everyday advice. It features a 2-way real-time voice assistant in 22+ Indian languages with live spoken word highlighting, an AI Sky Camera (Weather Lens) to predict rain from cloud photos, interactive 3D WebGL radar maps, and domain-specific advisories for farmers, travelers, and coastal fishermen."



Technology Stack

- **Frontend:** React 19, TypeScript, Vite, Tailwind CSS
- **3D Graphics:** Three.js (WebGL), THREE.Timer
- **Mapping:** Leaflet, OpenStreetMap, Nominatim Geocoding
- **Voice / Audio:** Web Speech API, HTML5 Audio, Web Audio
- **Backend:** Node.js / Express hosted on Render
- **AI Engine:** Google Gemini API for meteorological reasoning & vision



Top 5 Features to Highlight

- **1. 2-Way Voice Call:** Speak in Gujarati/Hindi/English; AI answers aloud with synced real-time word highlighting and auto-scrolling.
- **2. Weather Lens:** Snap a sky/cloud photo to get cloud type classification and instant rain risk %.
- **3. Kisan Advisory:** Crop spray windows, soil moisture & harvest warnings for farmers.
- **4. Interactive Radar:** Click any map coordinate to calculate rainfall in inches.
- **5. Travel Safety:** Transit hazard score between source and destination.



3 Critical Engineering Challenges Solved

- **Vercel 10s Timeout Fix:** Serverless functions timeout after 10s. We implemented direct client-side fetch from browser to Render backend (<https://weathergpt-backend-46or.onrender.com>), eliminating proxies and stream corruption.
- **Audio Lifecycle Management:** Added `stopAllAudio()` with refs to pause, reset `currentTime=0`, revoke Object URLs, and cancel Web Speech immediately on modal close/unmount.
- **User Privacy & History Isolation:** Scoped all chat sessions by authenticated Gmail (`weathergpt_sessions_v3_<email>`) so switching Google accounts completely isolates history.

Top Viva Questions & Instant Answers

Q1: How is WeatherGPT different from normal weather apps?

Ans: Normal apps show raw data (e.g. 70% humidity); WeatherGPT explains what that means in plain language—telling farmers when to spray crops and travelers which highways to avoid.

Q2: How does the real-time spoken word highlighting work?

Ans: SpokenTextRenderer breaks text into word tokens. onboundary / timeupdate events track the active character index, apply an animated pill highlight, and call `scrollIntoView({ behavior: 'smooth', block: 'center' })`.

Q3: How does Weather Lens analyze cloud images?

Ans: It encodes the captured photo to Base64, sends it directly to our Vision AI endpoint on Render, and receives cloud classification (Cumulonimbus/Cirrus) and rain risk percentage.

Q4: How did you implement multi-lingual support?

Ans: We created localized JSON schema dictionaries for all 22 official Indian languages and dynamically configure speech synthesis utterance languages (e.g. gu-IN, hi-IN, en-IN).

Live Demo: weather-gpt-tau.vercel.app | **GitHub:** github.com/sharaddetroja/WeatherGPT-